

Chapter 6: Automated and Emerging Technologies

Question 1: Compare industrial robots and domestic robots, highlighting their typical applications.

Answer: Industrial robots are used in manufacturing and assembly lines, while domestic robots perform tasks in households, such as vacuuming or lawn mowing.

Question 2: Define "automation" and provide an example of an automated process.

Answer: Automation is the use of technology to perform tasks with minimal human intervention. An example is an assembly line in car manufacturing.

Question 3: Describe the concept of "Machine Learning" and its role in autonomous systems.

Answer: Machine learning is a subset of artificial intelligence that enables machines to learn from data and make decisions without explicit programming. It is essential for autonomous systems like self-driving cars.

Question 4: Explain how Internet of Things (IoT) devices can enhance home automation.

Answer: IoT devices can connect and communicate with each other, enabling remote control and automation of various home functions like thermostats, lights, and security systems.

Question 5: Give an example of a wearable technology device and discuss its potential benefits.

Answer: A smartwatch is a wearable device that can track fitness, receive notifications, and provide health data, promoting a healthier lifestyle.

Question 6: Identify the key components of a robotic system, including sensors and actuators.

Answer: Key components include sensors (for input), actuators (for output), a controller, and power source.

Question 7: Suggest ways in which automation can improve efficiency in a manufacturing plant.

Answer: Automation can reduce human errors, increase production speed, and lower labour costs, improving overall efficiency.

Question 8: Define "Artificial Intelligence" (AI) and provide examples of AI applications.

Answer: AI is the simulation of human intelligence in machines. Examples include virtual assistants (e.g., Siri), recommendation systems (e.g., Netflix), and autonomous vehicles.